

Relationship between comfort and back posture and mobility in sitting-posture

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Abstract

The objective of this work is to analyze the causes of lumbar discomfort while sitting on a chair, by analyzing the relationship of lumbar curvature, pelvic inclination and their mobilities with discomfort. An experiment has been performed with healthy subjects, in which comfort, postural and mobility parameters have been measured. Their relationship has been analyzed with multivariate analysis. Factorial analysis has been used to represent all the correlated variables measured. Logistic regression and discriminant analyses have been used to classify discomfort/absence of discomfort. The results show that great changes of posture are a good indicator of discomfort, and that lordotic postures with forward leaned pelvis and low mobility are the principal causes of the increase of discomfort. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

In the evaluation of chairs, different methods have been used to measure comfort: anthropometry, subjective assessment and objective measurements such as postural, biomechanical and physiological parameters. Anthropometry is useful to define most functional dimensions of chairs, but it does not help with other features affecting comfort such as shape or inclination of seat pan and backrest (Bishu et al., 1991).

Subjective assessment from users is the only way to explore user preferences and detect changes in comfort and pain. However, these tests only provide qualitative information, so that their application is limited to comparing different models of chairs when they are evaluated simultaneously by the same group of subjects (Helander and Mukund, 1991). Several methods have been proposed to overcome this disadvantage: to establish admissible levels of discomfort from a database of results from many similar chairs (Drury and Coury, 1982), or to analyze the relationship between local pains

and general comfort, and to focus on reducing pain at the parts of the body that are most influential on comfort, by studying the origin of this pain (Bishu et al., 1991; Thomas et al., 1991; Tortosa et al., 1994). For sitting posture, lumbar pain has been shown to be the most important.

Objective measurements of postures and other biomechanical and physiological factors have been also widely used to analyze their relationship with different chair features (Anderson et al., 1979; Mandal, 1986; Nordin et al., 1986; Otun and Anderson, 1988; etc.). The aim of all these studies has been to examine what features contribute to minimize one particular factor, according to a predefined healthy posture, generally from a medical outlook. Sometimes in these studies, other factors are overlooked, temporal aspects are ignored, and/or chair features are analyzed independently or with chair-like prototypes. In addition, in some aspects, there are slightly opposing views about what the ideal posture should be. For instance, lumbar curvature should be as lordotic as possible as in standing posture or should be somewhat flexed.

However, subjective and objective parameters are not independent, so that the postures or the values of other physiological parameters that generate discomfort could

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be determined by studying their relationship. Few studies have been made to relate comfort/pain on a chair with objective measurements (Gross et al., 1994; Bishu et al., 1991; Genaidy and Karwowski, 1993; Yamazaki, 1992). Bishu et al. (1991) have shown that lumbar pain is related to lumbar curvature, although they have ignored temporal and mobility factors.

The aim of this work is to study in greater depth the causes of discomfort in lumbar region while sitting on a chair, by analyzing the relationship of discomfort not only with lumbar curvature but also with pelvic posture, the mobility of lumbar region and the use of the backrest.

2. Material and method

2.1. Measurement techniques

A *rachimeter* (Vergara and Page, 2000b) has been used to measure the posture in terms of lumbar curvature and pelvic inclination. The *rachimeter* (Fig. 1) contains a thin and flexible goniometer and a small inclinometer, and it allows the use of the backrest without affecting the spontaneous behavior of the subject and without the need for modifying the chair. The *rachimeter* is attached to the spine with a set of elastic straps and belts in such a way that the lumbar angle is measured from L5 to T12 spinous process levels. The small inclinometer on its base is located 1.5 cm under the start of the active zone of the goniometer, so that it measures sacrum inclination at 1.5 cm under L5 spinous process level, which may be used as an estimate of pelvic inclination (Moes, 1998).

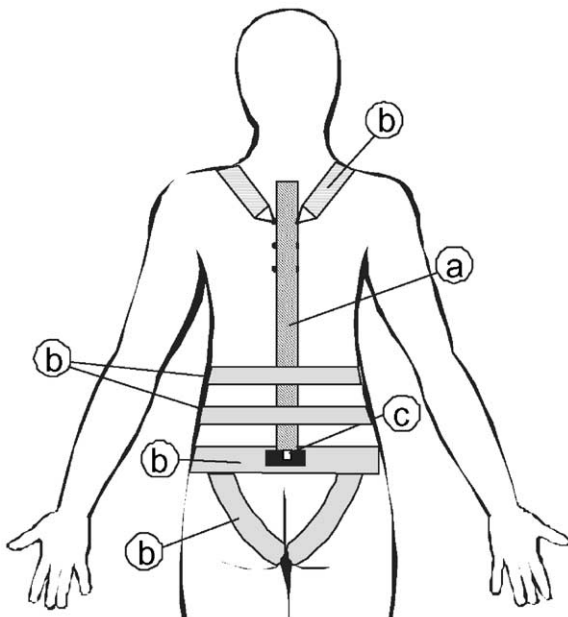


Fig. 1. Outline of the *rachimeter* and its attachment system: (a) *rachimeter*; (b) straps and belts and (c) base of the *rachimeter* and placement of the inclinometer on it.

Contact of the back with the backrest has also been measured with a device that allows detection of lumbar and/or dorsal contact with the backrest (Vergara and Page, 2000a). It consists of three flexible and thin electrodes (Fig. 2). One of them is attached to the backrest of the chair, another to the maximum lumbar concavity of subject's back and the last one to the maximum dorsal convexity. The subject wears adjusted clothing over the *rachimeter*, and the electrodes are attached on it with double-sided adhesive tape. Depending on the different combinations of contact between the three electrodes, the type of contact with the backrest is determined.

2.2. Experiment

An experiment was performed with 6 healthy subjects (3 males, 3 females), all of them volunteers. The subjects selected were different in size (height mean/standard deviation: 171/9.8 cm). Each subject used 6 different standard office chairs with a conventional office desk on different days, for a period of 100 min in each chair. During the test, the subjects were reading and writing.

The chairs (Fig. 3) were selected in order to present features as different as possible, especially in backrest size and shape, seat size and backrest/seat inclination. All of them were cushioned and with armrests, two of them presented no adjustable feature and the rest of them were adjustable at least in sitting height. The purpose of selecting quite different chair features and quite different subject sizes was to get comfort levels, postures and mobility as different as possible.

The posture and use of the backrest were measured in four 25-min consecutive periods, and at the end of every 25-min period, general comfort (GC) tests (Shackel et al., 1969) and body part discomfort (BPD) tests (Corlett and Bishop, 1976) were carried out. In the GC test, the subject is asked for his GC rate in a scale of 11 degrees (0: extreme discomfort, 10: extreme comfort). In the

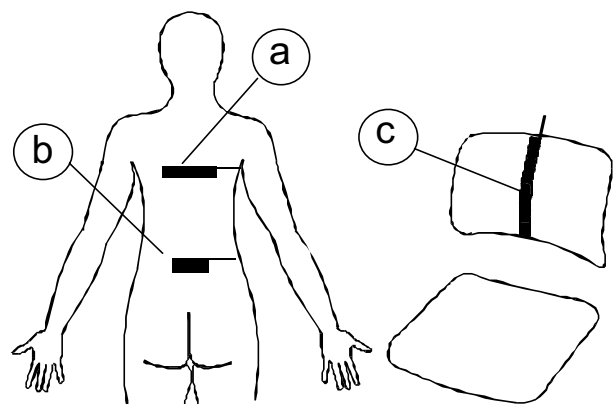


Fig. 2. Contact measurement system: (a) dorsal electrode, (b) lumbar electrode and (c) backrest electrode.



Fig. 3. Chairs used at the experiment.

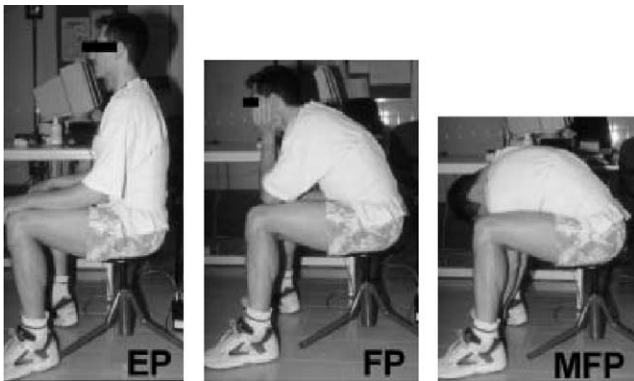


Fig. 4. Reference postures: EP—erect posture, FP—flexed posture with the elbows on the knees and the chin on the hands and MFP—maximum flexed posture with the arm between the legs trying to reach maximum flexion.

BPD test, a body part drawing is used and the subject is asked for the zones in which he feels the discomfort, and for these zones its intensity in a 5-point scale (1: light discomfort, 5: pain/extreme discomfort).

The subjects adjusted the features of each chair (when possible) to suit their preference and were requested to give spontaneous behavior, while remaining seated during the whole 100-min test. The order of subjects and chairs was randomized, but the hour of day each subject made the test was the same for all chairs.

In order to standardize lumbar angles for each subject, lumbar flexion and pelvic inclination were recorded at the beginning and the end of each test for three postures sitting on an adjustable stool with knees and ankles at 90° (Fig. 4): erect posture (EP), flexed posture (FP) with the elbows on the knees and the chin on the hands, and maximum flexed posture (MFP) with the arm between the legs trying to reach maximum flexion.

2.3. Data analysis

Firstly, the repeatability of the three possible postures for reference has been checked, both in the same session and in different sessions. Afterwards, the criteria for normalization have been selected to calculate parameters of mean postures. In order to do that, some ANOVAs

with factors ‘subject’ and ‘chair’ have been performed for the angles (lumbar and pelvic) without normalizing, and for different normalizations. The best normalization would be when the variance explained by the factor ‘subject’ is minimum, i.e. when the effect of the natural shape of the subject’s back is as small as possible.

Four parameters of mobility have been obtained for both lumbar and pelvic postures: amplitude and frequency of micro- and macro-movements. The angle records were filtered by means of digital signal filters to discriminate between both types of movements: fourth-order low pass butterworth digital filter with 1/90 Hz cut-off frequency for macro-movements and fourth-order high pass butterworth digital filter with 1/60 Hz cut-off frequency for micro-movements (to see an example, look at Vergara and Page, 2000a). In this way, macro-movements represent the global postural changes and micro-movements the small movements around postures maintained for some minutes. The parameters calculated were:

- *LRM/PRM*: Lumbar/pelvic range of motion in macro-movements. It represents the amplitude of the gross movements.
- *LFM/PFM*: Lumbar/pelvic frequency of motion in macro-movements, calculated as the number of postural changes greater than 5°.
- *LFA/PFA*: Lumbar/pelvic fidgeting amplitude, calculated as the rms of the micro-movements.
- *LFF/PFF*: Lumbar/pelvic fidgeting frequency, calculated as the zero-crossing times of the filtered signal of the micro-movements.

The percentage of time with different contact with the backrest has been calculated for each record:

- *LDC*: percentage of time with lumbar and dorsal contact;
- *DC*: percentage of time with dorsal contact;
- *LC*: percentage of time with lumbar contact;
- *NC*: percentage of time with no contact.

The correlation between postural and mobility parameters has been analyzed in terms of correlation coefficients. The BPD and GC relationship has been also analyzed to confirm the influence of local pain on comfort.

Multivariate analysis has been used to study the relationship of lumbar/pelvic posture and mobility with discomfort. Mean postures and micro-movements have been investigated as the cause of discomfort changes, and macro-movements have been investigated as objective indicators of discomfort. Factorial analyses have been used to represent all the postural variables. Logistic regression and discriminant analysis have been used to predict discomfort from the postural and mobility variables.

3. Results and discussion

3.1. Reference postures and normalization

The results of the repeatability of reference postures within the same session show that there are no significant differences between postures, except for the EP, in which the angles at the end of the tests are more flexed, possibly due to the tiredness of the subjects. The repeatability errors have been estimated as the mean residual standard deviation from an ANOVA with the factor 'subject' (Vergara and Page, 2000b). The bigger errors between sessions are presented as well by the EP ($> 5^\circ$ for both lumbar and pelvic angles), and the errors of both FP and MFP are of similar magnitude (2.5° for lumbar angle and 3.1° for pelvic angle). Therefore, the EP is discarded for normalizing, as its repeatability was not so good, and only ANOVAs for the angles without normalizing and for the differences with FP and MFP angles have been performed. The variances explained by the factor 'subject' are shown in Table 1. The purpose of normalizing the angles is to eliminate the effect of the subjects on the results to compare results with a minimum of reliability, therefore the best way of normalizing is that in which the variance explained by the factor 'subject' in these ANOVAs is minimum.

The difference with MFP has been used for normalizing lumbar angle as it represents a similar level of flexion for all the subjects, while the difference with FP depends on the length of the subject's limbs, and the variances are of similar magnitude. For pelvic angle, the angles without normalizing have been used.

Therefore, two parameters of mean postures have been calculated:

- *MLP*: Mean lumbar posture. Higher positive values indicate greater lordosis.
- *MPP*: Mean pelvic posture (vertical inclination of sacrum). Positive values mean forward bent pelvis.

3.2. Postural and mobility parameters

Table 2 shows the statistics of mean postures and mobility parameters for the 144 records of lumbar and pelvic angles. On average, the spontaneous lumbar mean

Table 2

Descriptive analysis of postural parameters. Angles in $^\circ$

	Mean	Maximum	Minimum	SD
MLP	12.5	27.3	-4.2	6.5
LRM	17.4	48.9	1.9	10.3
LFM	4.4	12	0	3.5
LFA	1.8	4.7	0.3	1.1
LFF	289	668	140	90
MPP	-22.8	-6.1	-51.1	8.5
PRM	20.9	70.5	1.6	14.4
PFM	4.0	14	0	3.4
PFA	1.6	5.4	0.1	0.9
PFF	327	603	123	118

posture is quite flexed, just 12° more lordotic than in MFP, and the pelvic mean posture is quite bent backwards ($> 22^\circ$). The movement range of the pelvis is bigger than the lumbar one, although the number of macro-movements of both regions is similar (a gross change of posture every 5 to 6 min as mean). The micro-movements of the lumbar zone are somewhat bigger and slower than for the pelvis.

All these parameters are quite correlated (Pearson's coefficient when possible, otherwise Kendall's coefficient): just 5 of the 45 possible relationships are statistically non-significant. The most important relationships follow. Lordotic lumbar posture is lightly associated with backward leaned pelvis. In general, when ranges of macro-movements are big, their frequencies are also high, and micro-movements have bigger amplitudes and smaller frequencies. Postures with lordotic lumbar curvature present less frequent movements of pelvis and quicker and smaller micro-movements. When the pelvis leans backward, pelvic and lumbar macro-movements are less frequent, while micro-movements are more frequent and with a slightly bigger amplitude.

The observed relationship of the use of the backrest with lumbar/pelvic mean posture and mobility is similar to that obtained in a previous experiment (Vergara and Page, 2000a).

3.3. Body part discomfort and general comfort

Although all the subjects used the same chairs, mean and initial GC levels are different for each subject ($p < 0.05$), but subjects do detect differences in comfort for the chairs ($p < 0.05$). Therefore, as it could be expected (Helander and Zhang, 1997), absolute levels of comfort are not comparable between subjects, but the relative mean comfort of each chair as well as the decrease/increase of GC levels measured in the same session may be compared for different subjects.

Table 1

Variances explained by the factor 'subject' of the ANOVAs with factors 'subject' and 'chair' for the lumbar and pelvic angles without normalizing and for the differences with FP and MFP angles

	Lumbar angle	Pelvic angle
Without normalizing	14663	5226
Difference with FP	2551	5832
Difference with MFP	2809	7107

Most frequent discomfort has appeared in the neck (37%), followed by lumbar zone (18%), and a little less in buttocks, dorsal zone and thighs (<10%). Other zone pains are present in <5% of the cases. This agrees with the results of larger studies on office work.

The correlation coefficients of BPD increase and GC decrease are only significant for lumbar zone ($\alpha < 0.01$) and neck and dorsal zone ($\alpha < 0.05$). The presence of lumbar discomfort is correlated as well with GC decrease ($\alpha < 0.05$). Therefore, lumbar pain is the most important pain, followed by neck and dorsal pain. Buttock and limb pain is not influential. This agrees with other authors' previous results (Bishu et al., 1991; Page et al., 1994).

3.4. General relationship between discomfort and posture/mobility parameters

Table 3 shows the association (Kruskal–Wallis test) of postural and mobility parameters with the increase of lumbar, neck and dorsal pains and the presence of lumbar pain.

In general terms, the increase of lumbar pain is associated with pelvic forward inclination, and slower and bigger micro-movements. In addition, when lumbar discomfort is present or increases, macro-movements are bigger and quicker. This fact strengthens the importance of lumbar pain, as it provokes more changes in posture, but also reinforces the idea that macro-movements are actually an effect of discomfort. Therefore, the answer to what postures induce changes in comfort must be explained from the mean postures and micro-movements parameters. Both relations, macro-movements as an effect of discomfort and mean postures and micro-movements as its cause, are analyzed next with multivariate analysis.

Table 3
Significant level of Kruskal–Wallis test for important discomforts and postural and mobility parameters^a

	Increase lumbar	Increase neck	Increase dorsal	Presence lumbar
MLP	○	+	—	○
MPP	+	○	+	++
LFA	○	○	○	+
LFF	—	○	○	—
PFA	+	○	○	++
PFF	○	○	○	—
LRM	○	+	○	○
LFM	++	○	○	++
PRM	++	+	○	++
PFM	+	○	+	++

^aThe sign +/— means an increase/decrease of the variable in the group with discomfort. +/— means $\alpha < 0.05$ (bilateral) + +/— means $\alpha < 0.01$ (bilateral)

3.5. Mean posture and micro-movements as the cause of discomfort

As these postural variables are highly correlated, their relationship with discomfort should not be analyzed independently with each variable, as it could give rise to false results. The variables should be previously reduced to a group of equivalent non-correlated variables with the use of a factorial analysis, and afterwards classification techniques such as logistic regression analysis could be used.

3.5.1. Factorial analysis to reduce correlated variables

Three factors (or new variables) may be used to represent the six original variables (more than 86% of the variance explained). The most significant components of each factor for the original variables are shown in Table 4.

The first component is correlated basically with the micro-movements variables. It measures roughly the *mobility*: increases with quick micro-movements (pelvic and lumbar) of low amplitude, and vice versa.

Both the second and third factors are correlated with mean lumbar and pelvic postures, but in opposite senses. The second factor is positive with lumbar kyphosis and pelvic forward inclination while the third factor is positive with lumbar lordosis and pelvic forward inclination.

The third factor represents the *anatomical relationship* described in the literature that lumbar kyphosis is associated with backward rotation of the pelvis (negative values of the factor), and lordosis is associated with forward rotation of the pelvis (positive values).

By analyzing the correlation of the second factor with the contact with the backrest it may be observed that positive values of the factor (kyphosis and forward pelvic inclination) correspond to periods in which the backrest is not used or when it is used as a support for the pelvis, i.e. when the work is done towards the table. On the contrary, negative values of factor 2 (lordosis with backward pelvic inclination) happen when the work is done reclining on the backrest. Therefore, the second factor could represent the *working habits* of

Table 4
Component matrix for the factorial analysis

	Components of factors		
	1	2	3
MLP		−0.583	0.697
LFA	−0.850		
LFF	0.877		
MPP		0.788	0.526
PFA	−0.863		
PFF	0.901		

the subject, with more support in the superior part of the back, or with the trunk bending forwards.

3.5.2. Logistic regression analysis to classify discomfort periods

In order to classify the cases with/without discomfort in the different regions, some logistic regression models have been adjusted, using the three factors (normalized) as classification variables (see Appendix A for details). Since the number of cases is less for the groups with discomfort, the cases have been weighted for all the analysis.

3.5.2.1. Lumbar pain. More than 67% of the cases with/without increase in lumbar pain are correctly classified. Table 5 shows the results.

The most influential factor is the *anatomical factor*, but in the sense that the probability that lumbar pain appears is bigger when the pelvis is rotated forward and there is more lordosis at the lumbar spine. Another factor influencing lumbar pain is *mobility*: static postures provoke more pain, while small and quick movements alleviate it. This confirms that small movements around a posture help to reduce muscular strain. *Working habits* factor has a smaller influence on lumbar pain.

In order to analyze the effect of the backrest in lumbar pain, another logistic regression analysis has been developed by adding the variable NC previously normalized. Table 6 shows the results. More than 67% of the cases were correctly classified. The results for the three factors are similar to the previous analysis, but the use of the backrest prevents the increase of lumbar pain.

3.5.2.2. Dorsal pain. Table 7 shows the results of the logistic regression to classify groups with and without

Table 5

Lumbar model: prediction of lumbar pain with the three factors (mobility, working habits and anatomical) as predicting variables

Variable	Exp(B)	R
F1: Mobility	0.549	−0.189
F2: Working habits	0.722	−0.093
F3: Anatomical	2.650	0.302

Table 6

Lumbar model: prediction of lumbar pain with the three factors (mobility, working habits and anatomical) and the absence of contact with the backrest as predicting variables

Variable	Exp(B)	R
F1: Mobility	0.606	−0.137
F2: Working habits	0.672	−0.107
F3: Anatomical	2.710	0.306
NC normalized	1.226	0.050

Table 7

Dorsal model: prediction of dorsal pain with the three factors (mobility, working habits and anatomical) as predicting variables

Factor	Exp(B)	R
F1: Mobility	0.357	−0.264
F2: Working habits	5.006	0.302
F3: Anatomical	1.211	0.063

increase in dorsal pain. More than 72% of the cases were correctly classified.

The decisive factor for dorsal pain is *working habits*: when the work is done bending forward, the probability of dorsal pain is quite superior than when the work is done reclining on the backrest. Another factor influencing dorsal pain is *mobility*, because of the continuity of the spine, lumbar movements are accompanied with movements of the dorsal zone, so that dorsal static stress is also alleviated. Finally, the *anatomical factor* affects dorsal pain very little.

3.5.2.3. Neck pain. As could be expected, in the prediction of neck pain, the model obtained is not significant and the percentage of correctly classified cases is low.

3.6. Macro-movements like objective indicator of comfort

The presence of lumbar pain has been considered as a discomfort indicator. Macro-movement variables are highly correlated as well, so that a factorial analysis has been made before performing the logistic regression analysis.

One factor is enough to represent the four macro-movement variables (it explains more than 71% of the total variance). The correlation of the factor with the original variables of macro-movements is bigger than 0.8 in all the cases. This factor measures *great changes of posture*: the factor has positive values with wide and frequent changes either for lumbar or pelvic movements.

The logistic regression with this factor as predicting variable to classify the presence of lumbar pains is significant ($R = 0.330$ and $\text{Exp}(B) = 2.745$), so that great changes in postures are a good indicator of discomfort. More than 68% of the cases are correctly classified.

In order to avoid the measurement of the four correlated variables and simplify the application of this criterion, a new model may be adjusted with just one of the original variables, the one that better correlates with the factor (PFM). The model classifies correctly more than 69% of the cases, and the results are quite similar ($R = 0.312$ and $\text{Exp}(B) = 1.281$).

With only this variable, a discriminant analysis could also be made, since the hypotheses to be applied are

fulfilled, so that quantitative results may be derived. This analysis predicts the value of the variable that better discriminates the groups. The results show that the number of pelvic posture changes that better discriminates (more than 74% of the cases correctly classified) is 4.04, for the 25-min intervals. This means that when pelvic changes are made in time intervals of <5 min for the tasks involved in this test, the probability of being uncomfortable is bigger than 0.5.

4. General discussion

In order to investigate the causes of discomfort while sitting, an experiment has been performed in which different subjects have used different chairs for reading and writing. Lumbar and pelvic postures and movements have been measured, as well as contact with the backrest. Two kinds of movements have been analyzed (macro-movements, which represent gross changes of postures, and micro-movements around these 'stable' postures) and two parameters for each of them have been computed (amplitude and frequency of movements). GC and BPD tests have been carried out.

As in wider studies, the most common discomfort appears in the neck and lumbar regions, followed by buttocks, dorsal region and thighs. However, only the increase of discomfort in lumbar, dorsal and/or neck regions provokes decrease in GC, i.e. buttock and thigh discomforts are not important for general feelings, only spine regions are. Besides, the only presence of some discomfort in the lumbar region provokes in itself a decrease of GC, so that lumbar pain is the most important for comfort.

In general terms, it has been observed that when lumbar pain increases, the mean posture of the pelvis is bent forward and micro-movements are slow and big, and when lumbar pain is present, macro-movements are bigger and quicker. Therefore, macro-movements have been quantified as an effect of discomfort while mean postures and micro-movements as its cause.

As the variables used to measure mean postures and micro-movements (amplitude and frequency) of pelvic and lumbar regions are, in general, quite correlated, a factorial analysis has been carried out. Three independent factors have been extracted to represent the six original variables. The first factor, *mobility*, represents the micro-movements. Both the second and third factors correlate with the mean pelvic and lumbar postures. The second factor has been associated with the *working habits* of the subject: with the trunk resting on the backrest or bending forward. The third factor represents the *anatomical* relationship described by other authors: forward inclination of pelvis with lumbar lordosis. The fact that two independent factors include lumbar and pelvic mean postures in opposite senses implies (as

expected) that lumbar and pelvic postures are not independent, but their relationship is not absolute and is much more influenced by the working habit than by the anatomical factor, i.e. depends strongly on the working habit.

The relationship of these three independent factors with lumbar pain has been analyzed with a logistic regression analysis. The most influential factor is the *anatomical* factor followed by the *mobility* factor: when the pelvis is rotated forward and there is more lordosis and/or the postures are static the probability for lumbar pain to appear is greater.

This negative effect of lordosis agrees with the result described by Bishu et al. (1991), although it must be remarked that the test has been developed working on a table with office chairs and for healthy subjects. Other previous works have shown (Williams et al., 1991) that, for patients with chronic backaches, lumbar pain diminishes with lordotic postures. On the other hand, these results should be validated in a long-term test, as comfortable postures in short term may be harmful in long term (Lueder, 1983). It has been checked that this negative effect on comfort is due to the muscular effort needed to maintain the lordosis, as comfort is strongly associated with muscular strain rather than with other aspects such as intradiscal pressure or the imitation of the 'natural' spine curve.

The positive effect of small movements around a posture to reduce muscular strain has already been considered by chair manufacturers, who produce chairs with flexible backrests (Reinecke and Hazard, 1994).

For the case of macro-movements as an effect of discomfort, it has been checked that they are a good indicator, and the best discriminant variable is the number of pelvic changes, i.e. when the changes in pelvic posture are made in time intervals of <5 min, the probability of being uncomfortable is bigger than 0.5.

5. Conclusions

It has been confirmed that lumbar pain is the most influential on GC when sitting on a chair, followed by neck and dorsal pain.

By means of multivariate analysis, it has been shown that the most important factor in the increase of short-term lumbar pain is to adopt lordotic and forward pelvic mean postures. The mobility of postures and the contact with the backrest help to decrease lumbar pain, but they have a lesser effect. With regard to dorsal pain, working reclined on the backrest produces less pain. These facts corroborate the consideration that static muscular effort is the main cause of lumbar and dorsal pain in short term.

It has been shown that great changes in posture are a good indicator of comfort. If the average time interval

between two consecutive changes is <5 min for this particular task, the probability of having lumbar pain is high.

Appendix A

Regression logistic models predict the odds ratio of belonging to either group as

$$\frac{\text{prob}(\text{group})}{\text{prob}(\text{no group})} = e^Z = e^{B_0 + B_1 X_1 + \dots + B_n X_n} \\ = e^{B_0} e^{B_1 X_1} \dots e^{B_n X_n}.$$

The value of $\text{Exp}(Bi)$ represents the multiplier to calculate the odds ratio when the independent variable i is increased by 1. By normalizing the predicting variables (mean 0 and standard deviation 1), the values of the coefficients may be compared with each other.

The statistic R represents the partial contribution of each variable, varying from -1 to $+1$. Small absolute values for R indicate that the variable has a small partial contribution to the model.

If the number of cases of one group is much less for one of the groups, the cases should be weighted to detect differences, but in this case, the constant of the adjusted model is distorted and cannot be interpreted, so that only qualitative results may be derived.

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